

Exercise - Do My Data Need Feature Engineering?

Checklist for evaluating whether a dataset requires feature engineering before applying machine learning algorithms.

Instructions

Use the checklist below to evaluate whether your dataset requires feature engineering before applying machine learning algorithms. For each item, answer Yes, No, or Not sure, and briefly justify your answer based on your previous exploratory analysis.

The purpose of this exercise is to connect data collection, cleaning, exploratory data analysis, and hypothesis testing with practical decisions about model-ready data representation.

1. General Structure of the Dataset

Guiding idea: if your dataset contains only clean numerical variables in meaningful scales, the need for feature engineering may be limited. If it contains categories, dates, text, skewed distributions, or complex relationships, feature engineering is probably necessary.

Question	Answer	Evidence from your dataset
Does your dataset contain variables that are not immediately usable by a machine learning algorithm?	Yes ☒ No ☐ Not sure ☐	sentiment_clean, emotion_group_v2, _source_file são strings; published/timestamp são datas; title/summary/url são texto bruto.
Are there columns that represent dates, categories, texts, codes, or identifiers?	Yes ☒ No ☐ Not sure ☐	IDs: discussion_id, url. Datas: published, timestamp. Texto: title, summary. Categóricas: sentiment_clean, emotion_group_v2, _source_file.
Are some variables useful in theory, but poorly represented in their current form?	Yes ☒ No ☐ Not sure ☐	Raw published é menos útil que componentes (published_hour/dayofweek/month); key_emotion_raw tinha ~30 labels colapsados em 9.
Are there variables that may need to be combined, transformed, or simplified?	Yes ☒ No ☐ Not sure ☐	num_theories skew=+2.54; emotion_group_v2 com 2 classes raras (humor, skepticism, n=4 cada); temas como binários separados.

2. Numerical Features

Example: if income ranges from 1,000 to 100,000 and is highly right-skewed, log_income may be more useful than raw income.

Question	Answer	Possible action
Do numerical variables have very different scales?	Yes ☒ No ☐ Not sure ☐	summary_length 0-101, intensity_imputed 4-9, engagement_score_v2 2-6.2.
Are some numerical variables strongly skewed?	Yes ☒ No ☐ Not sure ☐	num_theories skew=+2.54 (leptokurtic +6.20); theme_music +4.56; Shapiro-Wilk rejects normality ($p < 0.001$).
Are there extreme values that may distort the analysis?	Yes ☒ No ☐ Not sure ☐	2 mild outliers in engagement_score_v2 (IQR rule); num_theories with values up to 4 against a median of 0.
Are some numerical variables difficult to interpret in their raw form?	Yes ☒ No ☐ Not sure ☐	Absolute counts hide density — create theory_density = num_theories/summary_length and theme_diversity.
Do scatter plots suggest nonlinear relationships?	Yes ☐ No ☒ Not sure ☐	intensity × engagement $r = +0.80 \approx \rho = +0.82$ (linear). Principal relationships are linear or zero.

3. Categorical Features

Example: city should usually not be encoded as 1, 2, 3, because this would falsely suggest an order among cities.

Question	Answer	Possible action
Does your dataset contain nominal categorical variables?	Yes ☒ No ☐ Not sure ☐	sentiment_clean (3), emotion_group_v2 (9), _source_file (3).
Does your dataset contain ordinal categorical variables?	Yes ☒ No ☐ Not sure ☐	intensity_imputed (0-10) and sentiment_encoded (-1/0/+1) are ordinals.
Do some categorical variables have many distinct categories?	Yes ☒ No ☐ Not sure ☐	key_emotion_raw had ~30 values; title was practically unique per line.
Are some categories too rare to be analytically useful?	Yes ☒ No ☐ Not sure ☐	humor (n=4, 2.1%) and skepticism (n=4, 2.1%) in emotion_group_v2.
Would numerical encoding introduce a false order among categories?	Yes ☒ No ☐ Not sure ☐	Mapping _source_file day1/day3/week1 as 1/3/7 would impose a non-existent order.

4. Date and Time Features

Example: a raw date such as 2026-05-06 may be less useful than month = 5, weekday = Wednesday, or is_weekend = 0.

Question	Answer	Possible action
Does your dataset contain dates or timestamps?	Yes ☒ No ☐ Not sure ☐	Published timestamp converted to datetime64 in main.py.
Is the time elapsed since an event more meaningful than the date itself?	Yes ☒ No ☐ Not sure ☐	days_since_first_post, crawl_lag_hours already created.
Are there seasonal, weekly, or cyclical patterns?	Yes ☒ No ☐ Not sure ☐	published_hour (0-23) and published_dayofweek (0-6) are cyclic. Sin/cos encoding applied.
Are previous values relevant to current outcomes?	Yes ☐ No ☒ Not sure ☐	Posts are independent observations, not a time series.

5. Text Features

Example: student comments, product reviews, article titles, or institutional descriptions can be transformed into numerical features before modeling.

Question	Answer	Possible action
Does your dataset contain open-ended text responses, comments, titles, descriptions, or reviews?	Yes ☒ No ☐ Not sure ☐	The title and summary are raw Reddit text.
Could text length, word count, or frequency of terms be informative?	Yes ☒ No ☐ Not sure ☐	summary_length and title_length already created in main.py.
Are specific words or expressions analytically relevant?	Yes ☒ No ☐ Not sure ☐	7 canonical themes already extracted as binary flags (theme_character_development, etc.).
Is semantic similarity between texts important?	Yes ☐ No ☐ Not sure ☒	It could be useful for clustering, but it's not central with n=187.
Are texts too noisy or inconsistent?	Yes ☐ No ☒ Not sure ☐	LLM (sentiment + themes) analysis has already normalized the text into structured fields.

6. Redundancy and Dimensionality

Example: if income, wealth_index, and consumption_score are strongly correlated, they may provide overlapping information.

Question	Answer	Possible action
Do some variables measure almost the same thing?	Yes ☒ No ☐ Not sure ☐	num_themes ↔ num_canonical_themes; intensity_clean ↔ intensity_imputed; 3 versions of engagement_score.
Are some variables highly correlated with each other?	Yes ☒ No ☐ Not sure ☐	Heatmap (Lecture 7): intensity × engagement r=+0.80; 10 pairs with `
Does your dataset have many columns compared to the number of observations?	Yes ☒ No ☐ Not sure ☐	n=187 with 30+ columns in cleaned_dataset_improved; 70+ if one-hot naive.
Did encoding create many sparse columns?	Yes ☒ No ☐ Not sure ☐	9-cat emotion → 9 sparse binary characters (mitigated by grouping rare characters into rare).
Are some variables theoretically irrelevant to your research question?	Yes ☒ No ☐ Not sure ☐	URL, title_freq, raw title/summary strings are not modeled.

7. Connection with Future Machine Learning Models

Guiding idea: feature engineering depends not only on the dataset, but also on the algorithm and analytical objective.

Question	Answer	Possible implication
Would you use kNN, SVM, k-Means, DBSCAN, or PCA?	Yes ☒ No ☐ Not sure ☐	Clustering of discussion archetypes → mandatory scaling.
Would you use linear or logistic regression?	Yes ☒ No ☐ Not sure ☐	Predict engagement; multicollinearity already checked.
Would you use decision trees?	Yes ☒ No ☐ Not sure ☐	A natural fit for interpretable predictions.
Would you use clustering?	Yes ☒ No ☐ Not sure ☐	Discovery of post types (e.g. “positive nostalgic” vs “theoretical frustration”).
Would you prioritize interpretability?	Yes ☒ No ☐ Not sure ☐	Prefer simpler transformations and feature selection over opaque extraction methods.

Final Decision

After completing the checklist, classify your dataset into one of the following situations.

Classification	Meaning
Low need for feature engineering	The dataset is already mostly numerical, clean, interpretable, and model-ready.
Moderate need for feature engineering	Some variables require encoding, scaling, transformation, or simplification.
High need for feature engineering	The dataset contains many categorical, textual, temporal, skewed, redundant, or high-dimensional variables.

Short Written Answer

At the end, write a short paragraph using the template below.

Based on our exploratory analysis, our dataset requires [low/moderate/high] feature engineering. The main reasons are [...]. The most relevant operations would be [...], because [...]. Before applying machine learning algorithms, we should prioritize [...].

Notes

Based on our exploratory analysis, our dataset requires moderate feature engineering. The main reasons are the strong skewness in some numerical variables (num_theories), the presence of categorical variables that still require numerical encoding, and multicollinearity between related features, with 10 highly correlated pairs ($|r| \geq 0.9$). The most relevant operations would be log transformation, standardization, one-hot encoding, and feature selection, because they reduce skewness, align feature scales, convert categorical data into usable numerical representations, and remove redundant information. Before applying machine learning algorithms, we should prioritize encoding categorical variables, scaling numerical features, and selecting non-redundant features, especially given the modest sample size ($n=187$).